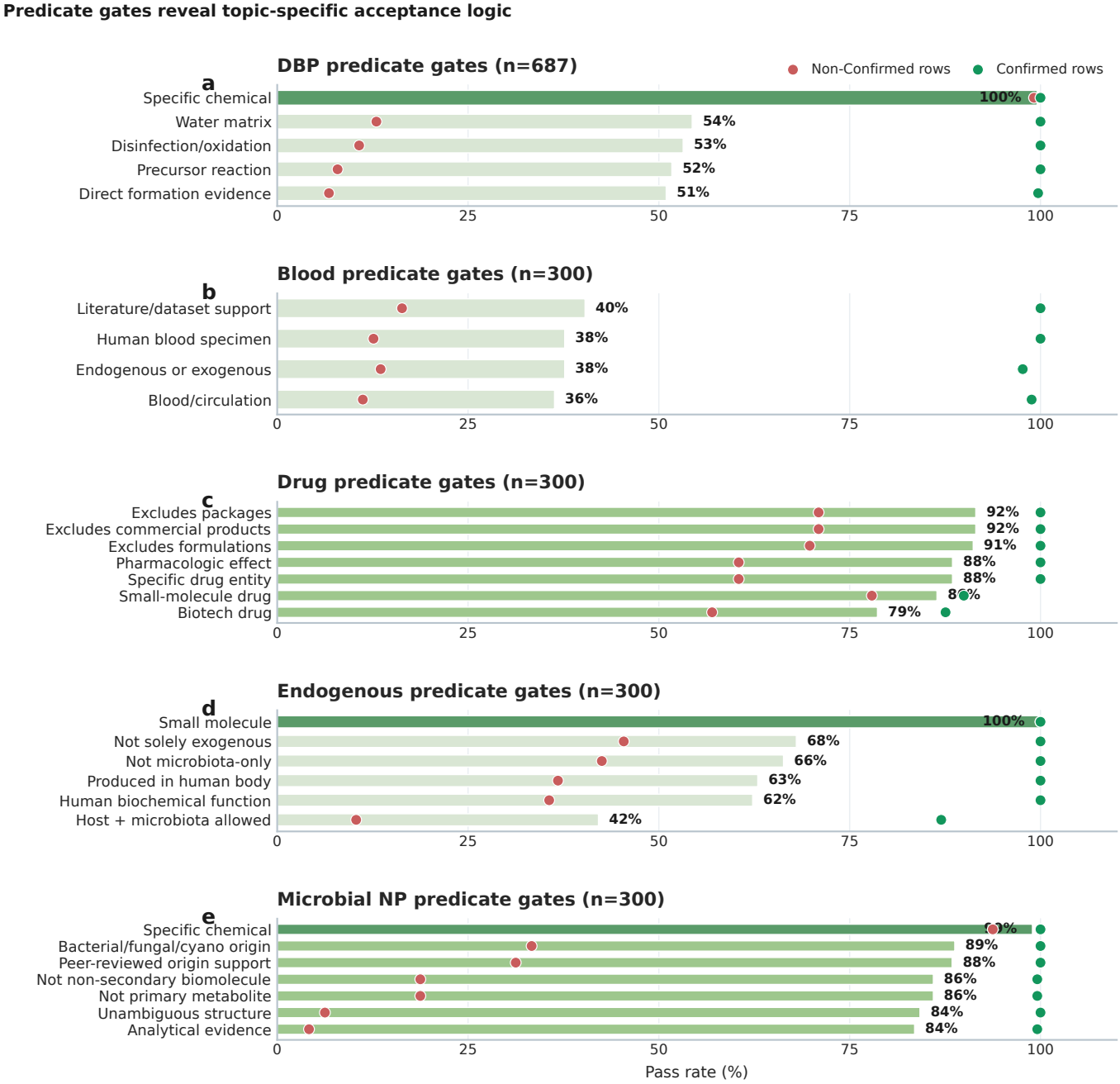
Predicate gates reveal topic-specific acceptance logic
a
DBP predicate gates (n=687)
Non-Confirmed rows
Confirmed rows
Specific chemical
Water matrix
Disinfection/oxidation
Precursor reaction
Direct formation evidence
100%
54%
53%
52%
51%
0
25
50
75
100
b
Blood predicate gates (n=300)
Literature/dataset support
Human blood specimen
Endogenous or exogenous
Blood/circulation
40%
38%
38%
36%
0
25
50
75
100
c
Drug predicate gates (n=300)
Excludes packages
Excludes commercial products
Excludes formulations
Pharmacologic effect
Specific drug entity
Small-molecule drug
Biotech drug
92%
92%
91%
88%
88%
79%
0
25
50
75
100
d
Endogenous predicate gates (n=300)
Small molecule
Not solely exogenous
Not microbiota-only
Produced in human body
Human biochemical function
Host + microbiota allowed
100%
68%
66%
63%
62%
42%
0
25
50
75
100
e
Microbial NP predicate gates (n=300)
Specific chemical
Bacterial/fungal/cyano origin
Peer-reviewed origin support
Not non-secondary biomolecule
Not primary metabolite
Unambiguous structure
Analytical evidence
89%
88%
86%
86%
84%
84%
0
25
50
75
100
Pass rate (%)